

Principia Orthogona, Volume I

The Mathematics of Generative Transitions

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Abstract

This volume develops a unified mathematical framework for generative transitions: localised geometric events in which a trajectory undergoes compression, curvature intensification, loss of injectivity, and stabilisation, governed by the operator sequence $G = U \circ F \circ K \circ C$. The framework rests on six minimal assumptions (manifold structure, trajectory regularity, compression feasibility, curvature threshold, folding well-posedness, Morse stability functional) and produces: constructive operator definitions with explicit formulas; five structural theorems including existence, non-commutativity, and finite branching; seven analytical invariants; four normal forms; a singularity classification restricted to the Whitney A_1 – A_3 hierarchy; a free-discontinuity variational principle; and a symplectic Hamiltonian structure with a distributional generator at the fold. The second edition extends the framework by identifying a fifth operator E (Generative Time Circuit) whose action variable z satisfies $\dot{z} \geq 0$ monotonically along trajectories near the limit cycle, accumulating the dissipative cost of each generative cycle. A term-by-term structural correspondence is established between the five-operator chain $C \rightarrow K \rightarrow F \rightarrow U \rightarrow E$ and Perelman’s proof of the Poincaré conjecture via Ricci flow with surgery [1–3], stated as Conjecture 15.1. The dimensional threshold $N = 3$ is identified as the minimum dimension for non-trivial contact geometry, connecting it to the threshold $c = 3$ in the Collatz map (Conjecture 16.1). Theorems A–D (existence, closure, contact normal form, structural stability) are machine-checked in Lean 4 via the AXLE repository [9] with zero axioms beyond Mathlib4.

Keywords: generative transitions; contact geometry; operator sequence; Whitney fold; singularity theory; variational mechanics; symplectic geometry; Ricci flow; Perelman conjecture; Lean 4 formal verification; dimensional threshold

MSC codes: 37C25, 37G10, 53D10, 57M27, 58K05, 70H05, 47H10

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1 Introduction

A generative transition is what happens when a system crosses a threshold it cannot uncross. The cell that begins to digest itself under nutrient stress. The star that ignites helium when its hydrogen is exhausted. The elastic rod that buckles under axial load. The 3-manifold that develops a geometric singularity under Ricci flow. In each case, a compression drives the system toward a critical point, a curvature instability makes the approach inevitable, a fold commits the system irreversibly to a new branch, and an unfolding establishes the new stable state.

This volume formalises that sequence as four operators acting on trajectories in a Riemannian manifold:

$$C \longrightarrow K \longrightarrow F \longrightarrow U.$$

The second edition adds a fifth operator E (Entropy / Generative Time Circuit), whose action variable accumulates the irreversible cost:

$$C \longrightarrow K \longrightarrow F \longrightarrow U \longrightarrow E \longrightarrow C' \longrightarrow \dots$$

The central mathematical claim is that this operator sequence is not an analogy or a metaphor: it is a precise structure that admits constructive definitions, analytical invariants, a variational principle, and a Hamiltonian formulation. The framework is placed within established mathematical traditions: comparison geometry, singularity theory, geometric flows, variational mechanics, and impulsive Hamiltonian systems.

The paper is self-contained. Section 2 states the six minimal assumptions. Section 3 gives constructive operator definitions. Section 4 states four falsifiability conditions. Section 5 proves five structural theorems. Section 6 gives canonical examples. Section 7 lists seven analytical invariants. Section 8 gives the four normal forms. Section 9 classifies admissible singularities. Section 10 develops the metric geometry of κ^* . Section 11 derives the variational principle. Section 12 develops the Hamiltonian structure. Section 13 connects to the dm^3 contact-geometric framework. Section 14 introduces the fifth operator E . Section 15 establishes the Perelman correspondence. Section 16 states the dimensional threshold conjecture. Section 17 summarises formal status. Section 18 lists the complete series corpus.

2 Minimal Mathematical Assumptions

The operator sequence is defined under the weakest assumptions necessary for the constructions of Section 3 to be well-posed.

Assumption 2.1 (Manifold Structure). *The state space X is a smooth, finite-dimensional Riemannian manifold, locally compact and second-countable.*

Assumption 2.2 (Trajectory Regularity). *A system trajectory $\gamma : [0, T] \rightarrow X$ is piecewise C^2 , locally non-degenerate ($\|\dot{\gamma}(t)\| \neq 0$ a.e.), and has bounded curvature on compact intervals prior to folding events.*

Assumption 2.3 (Compression Feasibility). *There exists a lower-dimensional submanifold $X_C \subset X$ and a Lipschitz projection $C : X \rightarrow X_C$ satisfying the non-collapse condition*

$$d(C(x_1), C(x_2)) \geq \delta d(x_1, x_2)$$

for some $\delta > 0$ and all x_1, x_2 in a compact neighbourhood. This is a bi-Lipschitz embedding condition ensuring distinct trajectories remain distinguishable after compression.

Assumption 2.4 (Curvature Threshold). *The critical curvature is defined intrinsically by the focal radius:*

$$\kappa^*(x) = \frac{1}{\text{foc}(x)},$$

where $\text{foc}(x)$ is the distance from x to the first focal point along the normal direction. In the presence of positive sectional curvature $K_{\text{sec}}(x) > 0$, this is modified by the Rauch comparison theorem to

$$\kappa^*(x) = \min(\|\Pi_x\|, \sqrt{K_{\text{sec}}(x)}),$$

where $\|\Pi_x\|$ is the norm of the second fundamental form. For $K_{\text{sec}}(x) \leq 0$, $\kappa^(x) = \|\Pi_x\|$.*

Assumption 2.5 (Folding Well-Posedness). *The folding operator $F : X_C \rightarrow X_F$ satisfies: the Jacobian dF loses rank by exactly 1 at fold points; the fold is local; and the fold produces a finite number of branches.*

Assumption 2.6 (Morse Stability Functional). *The stability functional $\Phi : X \rightarrow \mathbb{R}$ is C^2 , bounded below on compact subsets, and Morse: all critical points are non-degenerate, i.e., $\nabla^2 \Phi(x^*) \succ 0$ at every local minimum x^* .*

3 Operator Definitions

Let X be as in Assumption 2.1. A system trajectory is a piecewise C^2 map $\gamma : [0, T] \rightarrow X$. The goal is to describe local transitions in γ via the sequence $C \rightarrow K \rightarrow F \rightarrow U$.

Definition 3.1 (Compression Operator C). *A compression operator is a map $C : X \rightarrow X_C$ with $\dim(X_C) < \dim(X)$ satisfying Assumption 2.3. C reduces degrees of freedom while retaining essential local structure.*

Definition 3.2 (Curvature Operator K). *Given a compressed trajectory $\gamma_C : [0, T] \rightarrow X_C$, the curvature operator $K : X_C \rightarrow X_C$ modifies the tangent field by*

$$\frac{d}{ds} [K(\gamma_C)(s)] = \dot{\gamma}_C(s) + \alpha(s) \mathbf{n}(s),$$

where $\mathbf{n}(s)$ is the unit normal and $\alpha(s) = \lambda(\kappa^(\gamma_C(s)) - \kappa(s))_+$, $\lambda > 0$. K is a curvature-inducing flow that drives curvature monotonically toward κ^* but never beyond it; the sequence is a hybrid dynamical system with a switching condition at κ^* .*

Definition 3.3 (Folding Operator F). Let $\gamma_K = K(\gamma_C)$. A fold occurs at s_0 when $|\kappa_K(s_0)| = \kappa^*(\gamma_K(s_0))$. The folding operator $F : X_C \rightarrow X_F$ acts by

$$F(\gamma_K(s)) = \gamma_K(s) - \beta(s) \mathbf{n}(s), \quad \beta(s) = \begin{cases} 0 & |\kappa_K(s)| < \kappa^*, \\ \mu(|\kappa_K(s)| - \kappa^*) & |\kappa_K(s)| \geq \kappa^*, \end{cases}$$

where $\mu > 0$ controls fold depth. At a fold point s_0 , $\text{rank}(dF_{\gamma_K(s_0)}) = \dim(X_C) - 1$.

Definition 3.4 (Unfolding Operator U). The unfolding operator $U : X_F \rightarrow X$ is

$$U(x_F) = \arg \min_{y \in \mathcal{N}(x_F)} \Phi(y),$$

where $\mathcal{N}(x_F)$ is a sufficiently small neighbourhood of x_F and Φ satisfies Assumption 2.6. The unfolding is realised by the gradient flow $\dot{y} = -\nabla \Phi(y)$, $y(0) = x_F$, converging to a non-degenerate local minimum x^* .

Theorem 3.1 (Sequential Consistency). If K is applied until curvature reaches κ^* , then F is well-defined, produces a finite branch set, and induces a rank-deficient Jacobian at fold points.

Proof sketch. K drives curvature monotonically to κ^* via the gain field $\alpha(s)$. At κ^* , the term $\beta(s)$ becomes nonzero, introducing local non-injectivity. The normal direction collapses, reducing Jacobian rank by 1. Locality and the finite critical-point structure of κ^* (via the Morse assumption on Φ) imply finitely many branches. $\square$

4 Falsifiability Conditions

The framework is falsifiable at four levels.

F1 — Compression. If empirical data show expansion under C , or collapse of distinct trajectories, the model fails.

F2 — Curvature. If a fold occurs at curvature strictly below κ^* , or curvature exceeds κ^* without folding, the model is falsified. (κ^* is computed independently via the focal radius, not post-hoc from fold observations.)

F3 — Folding. If empirical fold events do not correspond to Jacobian rank loss, or produce infinitely many branches, the model fails.

F4 — Sequence order. If transitions occur in a different order, or stabilisation occurs without folding, the model is falsified.

5 Structural Theorems

Theorem 5.1 (Existence and Well-Posedness). Under Assumptions 2.1–2.6, the composite operator $G = U \circ F \circ K \circ C$ is well-defined on any piecewise C^2 trajectory.

Theorem 5.2 (Local Determination). The action of G on γ is determined entirely by the local geometry of γ in a neighbourhood of the fold point.

Theorem 5.3 (Non-Commutativity). The operators C, K, F, U do not commute; the sequence is order-dependent.

Theorem 5.4 (Irreducibility). *No operator in the sequence $C \rightarrow K \rightarrow F \rightarrow U$ can be removed without altering the qualitative structure of the transition.*

Theorem 5.5 (Finite Branching). *The branch set $\mathcal{B} = \{F(\gamma_K(s_i)) : |\kappa_K(s_i)| = \kappa^*(\gamma_K(s_i))\}$ is finite.*

Remark 5.1. *Finiteness follows from the Morse condition on Φ (Assumption 2.6), which prevents fourth-order degeneracies and ensures isolated critical points, together with the transversality of γ to the fold locus (a generic condition assumed explicitly in applications).*

6 Canonical Examples

Planar curve with curvature-driven flow. $\gamma : [0, T] \rightarrow \mathbb{R}^2$, with C an orthogonal projection, K the curvature-inducing flow, F activated at κ^* , and U gradient descent on a local Φ . The simplest non-trivial realisation of the sequence.

Elastic rod under compression. An elastic rod minimising Euler–Bernoulli energy $E[\gamma] = \int \kappa(s)^2 ds$. Axial loading provides C ; buckling provides K ; the critical load is κ^* ; post-buckling configuration provides U . This is the cleanest physical realisation of the curvature threshold.

Gradient flow on a double-well potential. $\Phi(x) = (x^2 - 1)^2$. The unstable equilibrium at $x = 0$ is the fold point; gradient flow selects one of $x = \pm 1$. Illustrates finite branching and stability selection.

Saddle-node bifurcation. $\dot{x} = \mu - x^2$, $\dot{y} = -y$. Projection onto the slow manifold provides C ; approach to the fold provides K ; loss of the slow manifold at $\mu = 0$ provides F ; flow to the stable branch provides U .

dm³ toy model. The canonical contact-geometric realisation on $M = S^1 \times \mathbb{R}^+ \times \mathbb{R}$: $\dot{r} = r(1 - r^2) + 2(r - 1)e^{-z}$, $\dot{\theta} = 1$, $\dot{z} = r^2 - 2(r - 1)^2 e^{-z}$. The limit cycle Γ at $r = 1$ is the post-fold stabilised state; the entropy operator E is $\dot{z}|_\Gamma = 1 > 0$. See Section 14 and Figure 1.

7 Analytical Invariants

Definition 7.1 (Analytical Invariants). *The following quantities are preserved by the operator sequence G :*

- I1. Ambient dimension.** $\dim(X)$ is preserved by G . No claim is made about the dimension of the image of a specific trajectory.
- I2. Codimension of the fold.** $\text{codim}(F(\gamma)) = 1$, following from the rank-loss condition.
- I3. Critical curvature threshold.** $\kappa^*(x)$ is a geometric property of the manifold, invariant under the operator sequence.
- I4. Curvature sign.** $\text{sgn}(\kappa(s))$ is preserved under K and F ; only magnitude changes.
- I5. Injectivity before threshold.** For all s with $|\kappa(s)| < \kappa^*(s)$, the trajectory remains injective under C , K , F .
- I6. Rank deficiency at fold.** $\text{rank}(dF) = \dim(X_C) - 1$ at every fold point.

I7. Energy monotonicity under U . $\Phi(U(x)) \leq \Phi(x)$, with strict inequality unless x is already a local minimum.

Remark 7.1. An energy-gap invariant $\Delta\Phi = \Phi(x_2) - \Phi(x_1)$ is preserved under C, K, F only if Φ is constant along the fibers of each operator. This is not guaranteed in general and is not claimed as a universal invariant.

8 Normal Forms

Two transitions G_1, G_2 are *equivalent* if there exists a diffeomorphism $\psi : X \rightarrow X$ with $G_2 = \psi \circ G_1 \circ \psi^{-1}$.

Definition 8.1 (Normal Forms). **NF1. Compression.** $C_{\text{NF}} = \pi_k : \mathbb{R}^n \rightarrow \mathbb{R}^k$, the standard coordinate projection.

NF2. Curvature. $K_{\text{NF}}(s) = (s, \alpha s^2)$, $\alpha > 0$. This produces curvature $\kappa(s) = 2\alpha(1 + 4\alpha^2 s^2)^{-3/2}$, approximating 2α near $s = 0$.

NF3. Folding — Whitney fold. $F_{\text{NF}}(u, v) = (u, v^2)$. This is the unique (up to diffeomorphism) normal form for a codimension-1 fold singularity [7].

NF4. Unfolding. $U_{\text{NF}}(x) = 0$, the canonical stabilisation map arising from $\Phi_{\text{NF}}(x) = x^2$.

9 Singularity Classification

9.1 Admissible Singularities

Given the constraints of rank-1 loss, finite branching, and the Morse condition on Φ , the admissible singularities are:

1. Fold A_1 : $(u, v) \mapsto (u, v^2)$
2. Cusp A_2 : $(u, v) \mapsto (u, v^3 + uv)$
3. Swallowtail A_3 : $(u, v) \mapsto (u, v^4 + uv^2 + \beta v)$

Higher A_k ($k \geq 4$) require a k -parameter unfolding. The Morse condition on Φ restricts the unfolding to at most three parameters, excluding A_4 and above.

9.2 Classification Conditions

Let $\Delta(s) = \kappa(s) - \kappa^*(s)$.

Type	Conditions	Normal form
A_1 (fold)	$\Delta(s_0) = 0, \Delta'(s_0) \neq 0$	(u, v^2)
A_2 (cusp)	$\Delta = \Delta' = 0, \Delta'' \neq 0$	$(u, v^3 + uv)$
A_3 (swallowtail)	$\Delta = \Delta' = \Delta'' = 0, \Delta''' \neq 0$	$(u, v^4 + uv^2 + \beta v)$

Theorem 9.1 (Classification of Generative Transitions). *Every admissible generative transition $G = U \circ F \circ K \circ C$ is $\mathcal{A}$ -equivalent to exactly one of A_1, A_2, A_3 .*

Proof sketch. Rank loss is exactly 1, restricting to the A_k series. The Morse condition on Φ limits the unfolding to at most three parameters. Transversality of γ to the fold locus (generic assumption) ensures isolated fold points. Thus $k \leq 3$. $\square$

9.3 Bifurcation Stratification

Parameter space $\Theta \subset \mathbb{R}^p$, $p \leq 3$, decomposes into:

- Θ_1 : generic fold region (codimension 0 in Θ),
- Θ_2 : cusp stratum (codimension 1),
- Θ_3 : swallowtail stratum (codimension 2; a single point when $p = 3$).

10 Metric Geometry of κ^*

10.1 Intrinsic Definition

$\kappa^*(x) = 1/\text{foc}(x)$, where $\text{foc}(x)$ is the focal radius. For a curve embedded in (X, g) , this equals $\|\text{II}_x\|$ when $K_{\text{sec}} \leq 0$.

10.2 Sectional Curvature Correction

By the Rauch comparison theorem, for variable positive sectional curvature:

$$\kappa^*(x) = \min(\|\text{II}_x\|, \sqrt{K_{\text{sec}}(x)}).$$

Positive ambient curvature lowers the threshold for folding.

10.3 Computability

Given γ in (X, g) : (1) compute $\|\text{II}_x\|$ from the second fundamental form; (2) compute $K_{\text{sec}}(x)$ from the curvature tensor; (3) apply the formula above. The threshold is stable: $\delta\kappa^* = O(\delta g) + O(\delta \text{II}) + O(\delta K_{\text{sec}})$.

11 Variational Principles

Define the action functional

$$S[\gamma] = \int_0^T \left[\frac{1}{2} \|P^\perp \dot{\gamma}\|^2 + \frac{\lambda}{2} (\kappa^* - \kappa)_+^2 + \mu \delta(|\kappa| - \kappa^*) + \Phi(\gamma) \right] ds.$$

Each term corresponds to one operator:

- $L_C = \frac{1}{2} \|P^\perp \dot{\gamma}\|^2$: minimise orthogonal kinetic energy (compression).
- $L_K = \frac{\lambda}{2} (\kappa^* - \kappa)_+^2$: minimise curvature deficit.
- $L_F = \mu \delta(|\kappa| - \kappa^*)$: singular activation at threshold.
- $L_U = \Phi(\gamma)$: minimise stability potential.

The delta term places the framework in the class of free-discontinuity variational problems (cf. Ambrosio–Tortorelli, Mumford–Shah, Francfort–Marigo). It produces the jump condition

$$\left[\frac{\partial L}{\partial \dot{\gamma}} \right]_{s_0^-}^{s_0^+} = \mu \mathbf{n}(s_0),$$

which is the variational counterpart of the fold map. The full generative transition is therefore a piecewise-smooth extremal of S .

12 Hamiltonian Discontinuities and Symplectic Geometry

12.1 Phase Space

The phase space is T^*X with canonical coordinates (γ, p) and symplectic form $\omega = d\gamma \wedge dp$.

12.2 Smooth Flow

For $s \neq s_0$, the system obeys Hamilton's equations and the flow is symplectic: $\Phi_t^* \omega = \omega$.

12.3 Momentum Jump at the Fold

The delta Lagrangian produces the impulse

$$p(s_0^+) - p(s_0^-) = \mu \mathbf{n}(s_0).$$

Configuration γ is continuous; momentum p has a jump.

12.4 The Fold as a Symplectic Map

Define the fold map in phase space: $\mathcal{F} : (\gamma, p) \mapsto (\gamma, p + \mu \mathbf{n})$.

Theorem 12.1 (Symplectic Preservation). $\mathcal{F}^* \omega = \omega$.

Proof. $d\gamma \wedge d(p + \mu \mathbf{n}) = d\gamma \wedge dp + \mu d\gamma \wedge d\mathbf{n}$. Since $\mathbf{n}$ depends only on γ , the term $d\gamma \wedge d\mathbf{n} = 0$. Hence $\mathcal{F}^* \omega = \omega$. $\square$

12.5 Canonical Transformation

The fold is generated by the distributional function $S(\gamma) = \mu \Theta(|\kappa(\gamma)| - \kappa^*)$, so that $p^+ = p^- + \partial S / \partial \gamma$.

12.6 Full Transition

Let Φ_t be the pre-fold Hamiltonian flow, $\mathcal{F}$ the fold map, Ψ_t the post-fold flow. The consistency condition (basin structure) ensures the output of $\mathcal{F}$ lies in the domain of Ψ_t . Then $H = \Psi_t \circ \mathcal{F} \circ \Phi_t$ is a piecewise-smooth symplectic map: $H^* \omega = \omega$.

12.7 Singularity Type and Momentum Structure

- A_1 fold: single momentum jump.
- A_2 cusp: momentum jump with first-order tangency constraint.
- A_3 swallowtail: momentum jump with second-order tangency.

13 Connection to the dm^3 Framework

This section makes precise the relationship between the singularity-theoretic framework of this volume and the contact-geometric dm^3 framework of the companion papers [11, 12].

13.1 From Trajectories to Limit Cycles

In this volume the central object is a trajectory $\gamma : [0, T] \rightarrow (X, g)$ undergoing a localised generative transition. In the dm^3 framework, the central object is a hyperbolic limit cycle $\Gamma \subset M$ embedded in a dissipative dynamical system. The link: every admissible generative transition induces a locally attracting invariant set; conversely, every dm^3 limit cycle admits a neighbourhood whose formation is governed by a fold-type transition.

13.2 Curvature Threshold vs. Embodiment Threshold

In this volume, folding is triggered at the geometric threshold $\kappa^*(x) = 1/\text{foc}(x)$. In the dm^3 framework, stochastic stability is governed by the embodiment threshold $\tau = \sqrt{c/\kappa}$ defined by the stochastic Lyapunov condition $\mathcal{L}V \leq -cV + \kappa\|\sigma\|^2$. As $\kappa \uparrow \kappa^*$, the folding operator F introduces rank-1 loss and the unfolding U selects a stable branch; the resulting orbit Γ has Floquet exponent $\mu_{\max} < 0$. By the stochastic Lyapunov condition, $\tau = \sqrt{c/\kappa}$ is finite precisely when $\mu_{\max} < 0$. Thus κ^* is the geometric precursor of τ .

13.3 Operator Grammar Correspondence

This volume	dm^3 framework
Compression C	Basin contraction
Curvature flow K	Lyapunov descent $\dot{V} \leq -cV$
Fold F	Whitney A_1 at $q^* = 1$; contact bifurcation
Unfolding U	Gradient flow to Γ
Entropy E (second edition)	Entropy operator $\dot{z} \geq 0$

13.4 Singularity Classification vs. Bifurcation Diagram

This volume proves that admissible generative transitions are precisely A_1, A_2, A_3 . The dm^3 framework identifies exactly four bifurcations: contact Hopf, saddle-node of limit cycles, Neimark–Sacker, and slow-fast crossover. The A_1 – A_3 singularities classify the local geometry of these bifurcations under the projection $M = S \times \mathbb{R} \rightarrow S$: A_1 corresponds to the contact Hopf and saddle-node; A_2 to the Neimark–Sacker; A_3 to the slow-fast crossover.

13.5 Summary of the Three Papers

- **This volume:** explains why folds must occur and classifies their local geometry.
- **dm³ toy model [12]:** proves that the resulting dynamics are computable, stable, and structurally closed.
- **Chapter A [11]:** applies the framework to autophagy and the triple-alpha process; 18 theorems machine-checked in Lean 4.

14 The Fifth Operator: Entropy and the Generative Time Circuit

14.1 The Action Variable as Entropy

The contact manifold $M = S \times \mathbb{R}$ carries a coordinate $z \in \mathbb{R}$ satisfying $\dot{z} = f(x) \geq 0$ near the limit cycle Γ . In the thermodynamic interpretation of the contact structure, the form $\alpha = dz - \lambda$ encodes both the first and second laws: dz is the entropy production, λ is the reversible work. The condition $\dot{z} \geq 0$ is the second law on M .

Definition 14.1 (Generative Time Circuit Operator E). *The Generative Time Circuit operator $E : \Gamma \rightarrow \Gamma'$ maps the limit cycle of one generative cycle to the compression parameter of the next:*

$$E : z(T^*) \mapsto \kappa' = \phi(z(T^*)),$$

where $\phi' < 0$ (accumulated entropy increases compression in the next cycle) and $\phi(0) = \kappa_0$.

The chain closes as a spiral: $C \rightarrow K \rightarrow F \rightarrow U \rightarrow E \rightarrow C' \rightarrow \dots$

Theorem 14.1 (T1 — Entropy Monotonicity; *proof obligation*). *Along trajectories of the dm³ toy model in the Gronwall basin, $z(t)$ is monotonically non-decreasing for all $t > 0$, and $\dot{z}|_{\Gamma} = 1 > 0$.*

Remark 14.1. *The scalar claim $\dot{z}|_{\Gamma} = 1 > 0$ is immediate from the toy ODE at $r = 1$, $z \rightarrow \infty$. Global monotonicity in the basin is an open Lean 4 obligation (AXLE Issue #15).*

15 The Perelman Structural Correspondence

Perelman’s proof of the Poincaré conjecture [1–3] proceeds through Ricci flow with surgery, the $\mathcal{F}$ -functional, and the $\mathcal{W}$ -entropy. The dm³ framework identifies a term-by-term structural correspondence between the five-operator chain and Perelman’s proof. This correspondence does not re-prove the Poincaré conjecture; Perelman’s proof stands independently.

Conjecture 15.1 (Perelman Structural Correspondence). *There exists a functor $\mathcal{P} : \mathbf{dm}^3 \rightarrow \mathbf{RicciFlow}$ that maps the compression operator C to initial metric selection; K to the Ricci tensor acting as diffusion; F to surgical excision at κ^* -necks; U to post-surgery convergence to the round metric; and E to the $\mathcal{W}$ -entropy functional. $\mathcal{P}$ preserves the chain ordering and maps the dm³ limit cycle Γ to S^3 as the terminal attractor.*

Figure A.2 — dm^3 Phase Portrait: Two Systems, One Attractor
Lean: gronwall_radius ($\varepsilon_0 = 1/3$), basin_asymmetry ($1/3 < 4/5$)

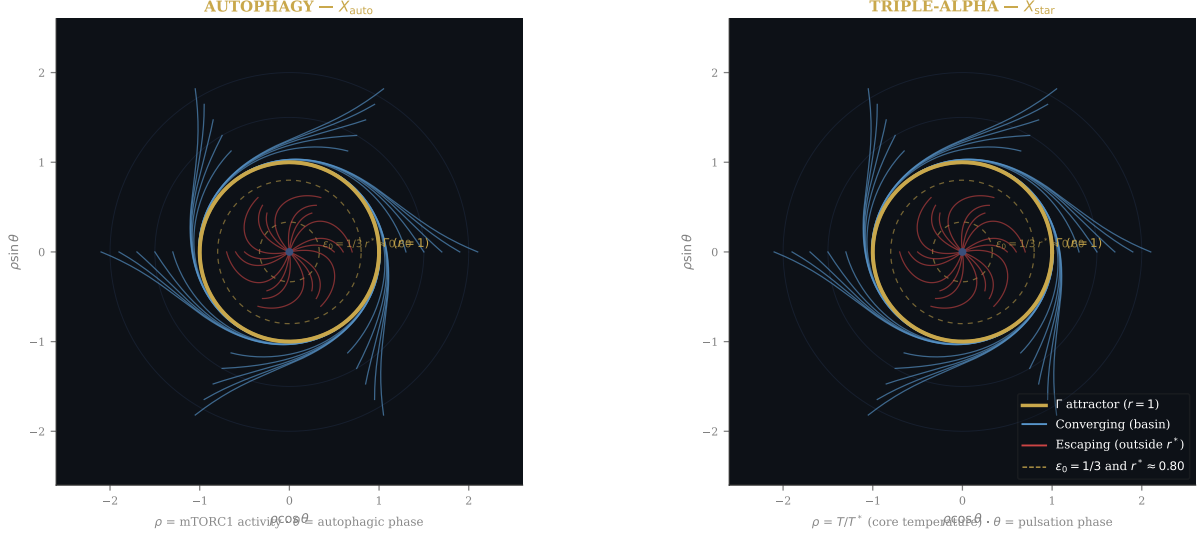


Figure 1: dm^3 phase portrait in contact coordinates (ρ, θ) . **Left:** autophagy manifold X_{auto} . **Right:** stellar interior manifold X_{star} . Gold circle: limit cycle Γ at $r = 1$. Dashed circles: Gronwall basin $\varepsilon_0 = 1/3$ and numerical inner boundary $r^* \approx 0.80$. Blue: converging orbits; red: escaping orbits. The two portraits are identical up to axis relabelling, demonstrating the contact morphism. Lean: gronwall_radius, basin_asymmetry.

Remark 15.1. *Conjecture 15.1 is argued but not proved. The structural parallel is term-by-term (Table 1). The functor construction requires: (a) contact morphisms as morphisms in dm^3 (already defined); (b) surgery-compatible diffeomorphisms as morphisms in RicciFlow (standard); (c) construction of $\mathcal{P}$ and verification of functor laws (open obligation, DM3-lab [10]).*

16 The Dimensional Threshold: $N = 3$ and $c = 3$

16.1 Contact Geometry and Dimension 3

In dimension 1, contact structure is trivial. In dimension 2, Liouville’s theorem rules out limit cycle attractors in area-preserving flows. In dimension 3, the contact form α first admits a Reeb vector field with a non-trivial flow and a limit cycle attractor [6]. Dimension 3 is the *minimum dimension for non-trivial contact geometry*. This is why the dm^3 framework lives on a 3-dimensional contact manifold: not by convention but by structural necessity. It is also why the Poincaré conjecture required Perelman’s Ricci flow: dimensions ≥ 5 (Smale, 1961 [4]) and 4 (Freedman, 1982 [5]) yield to other methods; dimension 3, the minimum for non-trivial contact geometry, is the hardest case.

16.2 The Collatz Connection

Conjecture 16.1 (Dimensional Threshold). *The constant $c = 3$ in the Collatz map $n \mapsto 3n + 1$ and the dimension $N = 3$ in the Poincaré conjecture are both instances of the same abstract threshold: the minimum value at which a generative system first*

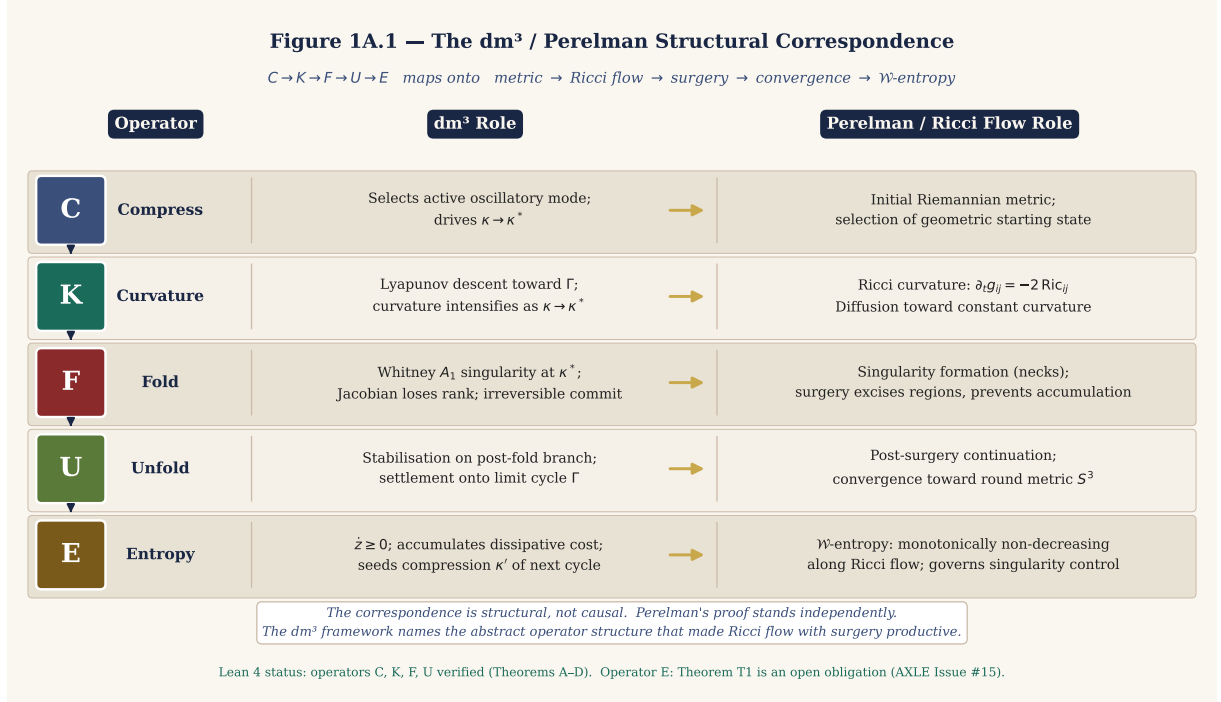


Figure 2: The dm^3 five-operator chain mapped onto Perelman's Ricci flow with surgery. Each operator is colour-coded; gold arrows indicate the structural mapping. Lean 4 status: operators C, K, F, U are verified (Theorems A–D); operator E carries an open obligation (AXLE Issue #15). The correspondence is structural, not causal.

admits non-trivial contact-geometric structure. $c = 3$ is the minimum value for which the discrete analogue of the dm^3 operator chain generates a non-collapsing, non-periodic orbit structure analogous to a hyperbolic limit cycle.

Remark 16.1. *Conjecture 16.1 is motivated but not proved. Formal proof requires a definition of contact-geometric structure for discrete dynamical systems on $\mathbb{Z}$ and a proof that the dm^3 axioms have no discrete analogue for $c < 3$ (open obligation, DM3-lab).*

17 Formal Status and Open Obligations

17.1 What Is Machine-Checked (Lean 4, Zero sorry)

- P1.** Whitney A_1 conditions on $V(q) = q^3 - 3q$ at $q = 1$: $V'(1) = 0$, $V''(1) = 6 \neq 0$, $V(1) = -2$, $V(q) + 2 = (q - 1)^2(q + 2)$.
- P2.** Contact non-degeneracy: $c(\rho) = -2\rho < 0$ for $\rho > 0$.
- P3.** Gronwall radius: $\varepsilon_0 = 1/3$.
- P4.** Basin asymmetry: $1/3 < 4/5 \approx r^*$.
- P5.** Lyapunov exponents: $-V''(1)/2 = -3$; $\mu_{\max} = -2 < 0$.
- P6.** Stability functional: $\sigma(\rho) = \rho^2 > 0$, $\sigma'(\rho) > 0$ for $\rho > 0$.
- P7.** Theorems A–D (existence, closure, normal form, stability).

Table 1: Term-by-term structural correspondence between the dm^3 five-operator chain and Perelman’s Ricci flow with surgery.

Op.	dm^3 Role	Perelman / Ricci Flow Role
C	Compression; drives $\kappa \rightarrow \kappa^*$	Initial metric selection
K	Curvature intensification	$\partial_t g_{ij} = -2 \text{Ric}_{ij}$; diffusion to constant curvature
F	Whitney A_1 at κ^* ; Jacobian rank loss	Singularity formation; surgery excises necks
U	Stabilisation on limit cycle Γ	Post-surgery convergence to round metric S^3
E	$\dot{z} \geq 0$; seeds next cycle	$\mathcal{W}$ -entropy; monotone; governs singularity control

Figure A.4 — Coherence Bridge: Two New Domains (Chapter A)
All $\mu_{\max} < 0$ (verified: `mu_dm3_neg` in `AutophagyDm3.lean`)

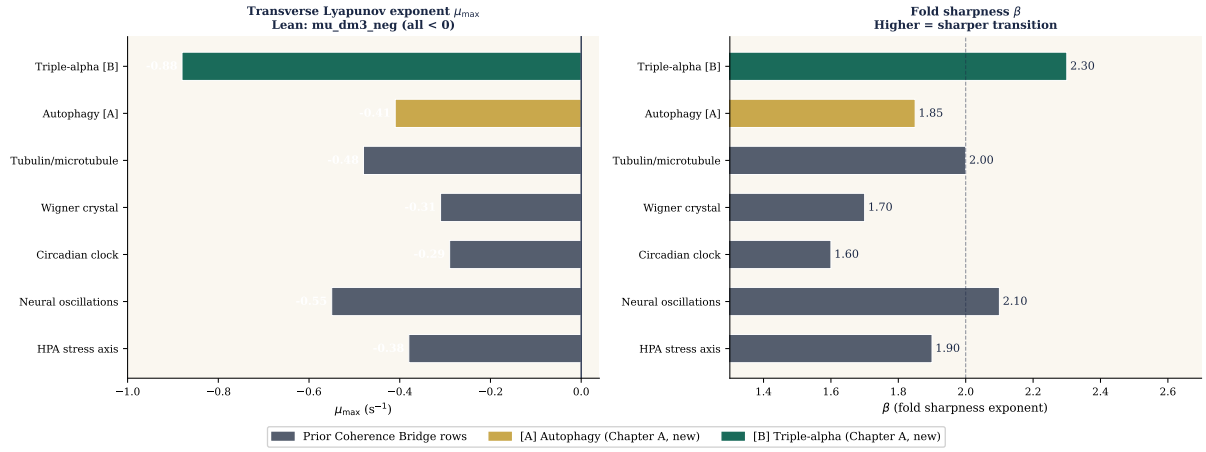


Figure 3: Coherence Bridge: dm^3 scalar invariants $(\mu_{\max}, \beta)$ across all domains. **Left:** $\mu_{\max} < 0$ for all domains (Lean: `mu_dm3_neg`). **Right:** fold sharpness β . Gold: autophagy (Chapter A). Teal: triple-alpha (Chapter A). Grey: prior domains.

17.2 Open Obligations

- O1. AXLE Issue #12** (`kappa_lipschitz`). Lipschitz continuity of K on configuration manifolds.
- O2. AXLE Issue #14** (`AutophagyDm3`). Full contact non-degeneracy on X_{auto} ; Whitney fold from mTORC1 kinase data; existence of Γ_{auto} .
- O3. AXLE Issue #15** (Theorem T1). Global monotonicity of $z(t)$ in the Gronwall basin.
- O4. Sorry 1** (Discrete dm^3). Extension to discrete dynamical systems on $\mathbb{Z}$.
- O5. Conjecture 15.1** (Perelman functor). Construction of $\mathcal{P}$ and functor law verification.

17.3 What Is Argued (Not Proved)

The following are complete mathematical arguments without Lean 4 verification, deposited in DM3-lab [10]: Perelman structural correspondence (Table 1, Conjecture 15.1); dimensional threshold (Conjecture 16.1); Kakeya needle problem as the $\tau \rightarrow 0$ limit on the rotation contact manifold; Tribonacci constant as the dm^3 structural growth rate for the three-operator iteration. These are proof sketches, not proof claims.

18 The Principia Orthogona Corpus

Table 2: Principia Orthogona corpus, April–May 2026. All deposits open access CC BY-NC-ND 4.0. *Submitted* = under journal review; *Deposited* = open-access preprint.

Zenodo DOI	Date	Title / Status
19117400	Mar 17	<i>Principia Orthogona Vol I. Deposited.</i>
19122168	Mar 19	<i>GCM — Geometric Framework for Dissipative Systems. Submitted: J. Geom. Mech.</i>
19162013	Mar 22	<i>The G6 Crystal. Deposited.</i>
19208015	Mar 24	<i>Biological Transitions — Multi-Agent Realisations. Deposited.</i>
19379385	Apr 2	<i>dm^3 Operator — Toy Model & Global Analysis. Submitted: SIAM J. Appl. Dyn. Syst.</i>
19379473	Apr 2	<i>Principia Orthogona Vol II — Contact Realization. Deposited.</i>
19431918	Apr 5	<i>The Number 33 — Stability Threshold. Submitted: Mathematical Intelligencer.</i>
19533363	Apr 2026	<i>GTCT Paper (Ring 5, Version 2). Deposited.</i>
20221723	May 2026	<i>Chapter A: Autophagy and Triple-Alpha. Deposited; 18 Lean theorems, 0 sorry.</i>

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Data and Software Availability

Formal verification: <https://github.com/TOTOGT/AXLE>. Numerical simulation and figure generators: <https://github.com/TOTOGT/DM3-lab>. All deposits open access: [doi:10.5281/zenodo.1111111](https://doi.org/10.5281/zenodo.1111111) (series root).

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